

RESEARCH PAPER / INTERNSHIP PROJECT REPORT

Machine Learning-Based Employee Attrition Prediction and Risk Scoring System

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Place: Panikhaiti, Guwahati, Assam

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DECLARATION

I, Samsul Hoque, hereby declare that the project report titled "Machine Learning-Based Employee Attrition Prediction and Risk Scoring System" has been prepared as part of my unified mentor - 3-month data analyst course / internship. The work presented in this report describes the analytical and machine-learning implementation carried out for the project. The reported model metrics are taken from the project's held-out test evaluation, and the report distinguishes project-generated results from information obtained from published literature.

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Signature: Samsul Hoque

ABSTRACT

Employee attrition is a major workforce-management challenge because unexpected employee turnover can affect continuity, productivity, recruitment workload, and organizational knowledge. This research presents a machine-learning-based Employee Attrition Prediction and Risk Scoring System developed as part of a three-month Data Analyst internship with Unified Mentor. The system uses an HR employee dataset containing 1,470 employee records and 31 project-level columns. The workflow includes data validation, exploratory analysis, feature engineering, categorical encoding, numerical scaling, class-imbalance treatment using SMOTE, model training, evaluation, risk scoring, and interactive visualization. Logistic Regression, Random Forest, and Gradient Boosting were trained and evaluated using an 80:20 stratified train-test split. On the held-out test set, Logistic Regression achieved 76.19% accuracy and 0.7999 ROC-AUC; Random Forest achieved 85.03% accuracy and 0.7961 ROC-AUC; and Gradient Boosting achieved 86.05% accuracy and 0.7929 ROC-AUC. The project also converts predicted probabilities into three operational risk categories: Low, Medium, and High. A Streamlit dashboard was developed to support workforce risk exploration, employee-level analysis, what-if scenarios, department and job-role analysis, model-performance review, and downloadable reports. The system is intended as an analytical decision-support prototype rather than a replacement for human HR judgment.

Keywords: Employee Attrition, Machine Learning, HR Analytics, Risk Scoring, SMOTE, Gradient Boosting, Random Forest, Streamlit, Predictive Analytics

1. INTRODUCTION

Employee attrition refers to employees leaving an organization through resignation, retirement, termination, or other forms of separation. For HR teams, understanding attrition patterns is important because workforce changes can create recruitment requirements, training costs, knowledge loss, and workload redistribution.

Traditional HR analysis is often retrospective: teams examine historical turnover after it occurs. Machine learning provides a complementary approach by learning patterns from historical employee data and estimating the probability that an employee belongs to an attrition-risk class. Research using the widely studied IBM HR Analytics benchmark has demonstrated that employee demographics, compensation, tenure, job characteristics, satisfaction, and work conditions can be useful predictive variables.

This project develops an end-to-end analytical system that combines supervised machine learning with risk scoring and an interactive dashboard. The focus is not only on model prediction but also on presenting results in a form that can be explored by non-technical users.

2. PROBLEM STATEMENT

Organizations may possess substantial employee information but still lack a practical mechanism for translating that information into an employee-level attrition risk indicator. A useful system should address several challenges simultaneously:

- identify patterns associated with historical employee attrition;
- handle an imbalanced target variable where employees who stay substantially outnumber employees who leave;
- preprocess mixed numerical and categorical HR attributes consistently;
- compare multiple machine-learning algorithms using appropriate evaluation metrics;
- transform model probabilities into understandable risk categories; and
- provide an interface through which HR-oriented users can inspect aggregate and individual results.

The proposed system addresses these requirements through a reproducible Python pipeline and a Streamlit-based dashboard.

3. OBJECTIVES

The major objectives of the project are:

1. Analyze the structure and quality of the employee HR dataset.
2. Prepare numerical and categorical variables for machine learning.
3. Engineer additional features representing income-to-experience, promotion delay, engagement, and overtime-related workload.
4. Address class imbalance during model training using SMOTE.
5. Train and compare Logistic Regression, Random Forest, and Gradient Boosting models.
6. Evaluate models using accuracy, precision, recall, F1-score, and ROC-AUC.
7. Generate an employee-level attrition probability and convert it into Low, Medium, or High risk.
8. Develop an interactive Streamlit dashboard for HR-oriented analysis.
9. Provide what-if analysis so that selected employee attributes can be changed and the predicted risk recalculated.
10. Produce a downloadable employee risk report for analytical use.

4. LITERATURE REVIEW

Employee attrition prediction has been widely studied using the IBM HR Analytics Employee Attrition & Performance benchmark. Recent literature continues to evaluate Logistic Regression, Random Forest, Gradient Boosting, XGBoost, SVM, and ensemble approaches, while also emphasizing class imbalance and explainability.

A 2022 Applied Sciences study used the IBM HR employee dataset containing 1,470 records and 35 features for employee attrition prediction. More recent studies have compared multiple classifiers and have explored SMOTE and explainable-AI techniques. A 2026 Scientific Reports study specifically highlighted the importance of evaluating generalizability and using explainability when deploying attrition models. A 2026 Information paper similarly investigated ensemble learning, feature engineering, and imbalance treatment for turnover prediction.

The present project follows this research direction while implementing a compact, reproducible pipeline tailored to the supplied project dataset and a practical Streamlit interface. Its contribution is the integration of preprocessing, feature engineering, balanced model training, probability-based risk scoring, what-if analysis, and dashboard reporting in a single workflow.

5. DATASET DESCRIPTION

The project dataset is stored locally as "Palo Alto Networks.csv". It contains 1,470 employee records and 31 columns used in the project workflow. The target variable is Attrition, represented as Yes/No in the source data and converted to a binary target for modelling.

The dataset contains demographic, compensation, job, satisfaction, tenure, travel, workload, and performance-related attributes. Representative variables include Age, BusinessTravel, Department, DistanceFromHome, Education, EnvironmentSatisfaction, JobInvolvement, JobLevel, JobRole, JobSatisfaction, MonthlyIncome, NumCompaniesWorked, OverTime, PerformanceRating, RelationshipSatisfaction, StockOptionLevel, TotalWorkingYears, TrainingTimesLastYear, WorkLifeBalance, YearsAtCompany, YearsInCurrentRole, YearsSinceLastPromotion, and YearsWithCurrManager.

The project-level dataset contains 1,233 employees recorded as not having left and 237 employees recorded as having left, corresponding to 83.88% and 16.12%, respectively. This imbalance is important because a model can obtain high overall accuracy while still missing a meaningful portion of the minority attrition class.

6. DATA PREPROCESSING

The preprocessing pipeline was designed to provide the same transformation logic during training and prediction.

First, the target column Attrition was separated from the predictor matrix. The remaining features were divided into categorical and numerical groups based on their data types. Numerical variables were standardized using `StandardScaler`. Categorical variables were converted using `OneHotEncoder` with `handle_unknown='ignore'`, allowing the model to process categories consistently even when an unseen category appears during prediction.

The transformed training data were then balanced using SMOTE. The training set originally contained 986 non-attrition observations and 190 attrition observations. After SMOTE, the two classes were balanced at 986 observations each. The test set was not synthetically oversampled and remained an unseen evaluation set.

An 80:20 stratified split was used, resulting in 1,176 training records and 294 test records. Stratification preserved the approximate class proportions between training and test subsets.

7. FEATURE ENGINEERING

Four project-specific features were engineered to add analytical signals beyond the original columns.

1. $\text{IncomeExperienceRatio} = \text{MonthlyIncome} / (\text{TotalWorkingYears} + 1)$

This represents monthly income relative to total working experience while avoiding division by zero.

2. $\text{PromotionDelay} = \text{YearsAtCompany} - \text{YearsSinceLastPromotion}$

This provides a simple measure related to time spent at the organization relative to the most recent promotion interval.

3. $\text{EngagementScore} = \text{JobInvolvement} + \text{JobSatisfaction} + \text{EnvironmentSatisfaction} + \text{RelationshipSatisfaction}$

This combines four ordinal workplace-experience indicators into a compact engagement-related feature.

4. $\text{WorkloadStressFlag} = 1$ when `OverTime` is Yes, otherwise 0

This converts the overtime condition into a machine-learning-friendly binary indicator.

These features were generated before preprocessing and were passed through the same transformation pipeline as the original predictors.

8. MACHINE LEARNING METHODOLOGY

Three supervised classification models were implemented.

Logistic Regression was included as an interpretable linear baseline and provides a probability-oriented classification approach.

Random Forest was selected as a tree-ensemble method capable of modelling non-linear relationships and interactions among HR variables.

Gradient Boosting was selected as another ensemble approach that sequentially improves weak learners to reduce prediction error.

All three models were trained using the same processed feature representation. Model evaluation was performed on the held-out test set using Accuracy, Precision, Recall, F1-score, and ROC-AUC. These metrics were considered together because attrition prediction is an imbalanced classification problem and accuracy alone is insufficient to describe minority-class performance.

9. EXPERIMENTAL RESULTS

The final held-out test results produced by the project training pipeline are shown below.

Logistic Regression:

- Accuracy: 0.7619 (76.19%)
- Precision: 0.3580 (35.80%)
- Recall: 0.6170 (61.70%)
- F1-score: 0.4531 (45.31%)
- ROC-AUC: 0.7999

Random Forest:

- Accuracy: 0.8503 (85.03%)
- Precision: 0.5600 (56.00%)
- Recall: 0.2979 (29.79%)
- F1-score: 0.3889 (38.89%)
- ROC-AUC: 0.7961

Gradient Boosting:

- Accuracy: 0.8605 (86.05%)
- Precision: 0.6071 (60.71%)
- Recall: 0.3617 (36.17%)
- F1-score: 0.4533 (45.33%)
- ROC-AUC: 0.7929

These values are test-set measurements from the project implementation. They should not be interpreted as proof that the model will perform identically on a different organization or future workforce dataset.

10. RISK SCORING FRAMEWORK

The system converts the Gradient Boosting model's predicted probability of attrition into a percentage risk score.

$\text{RiskScore} = \text{Predicted probability of Attrition} = \text{Yes} \times 100$

The project defines three operational categories:

- Low Risk: below 30%
- Medium Risk: 30% to 60%
- High Risk: above 60%

Using the full employee risk-report generation workflow, the project produced 1,173 Low Risk, 169 Medium Risk, and 128 High Risk records. These full-dataset probabilities are generated by applying the trained model to the available employee records; therefore, they are not equivalent to an independent validation estimate of future performance.

11. STREAMLIT DASHBOARD

An interactive Streamlit dashboard was developed as the application layer of the project. The interface separates presentation components from the main application controller through a frontend package containing styles, reusable components, and dashboard orchestration.

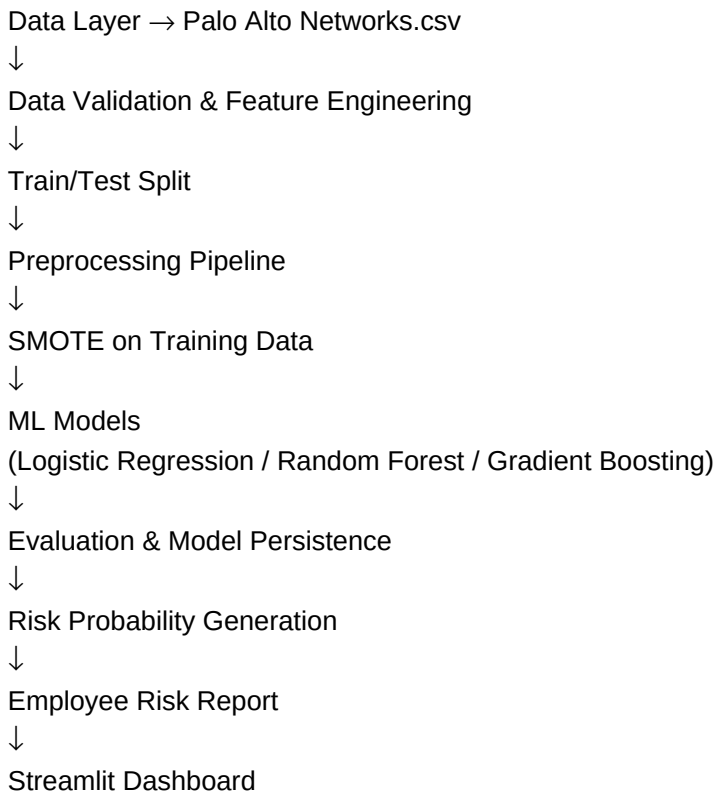
The dashboard includes:

- workforce risk KPIs;
- sidebar filtering;
- risk distribution and percentage analysis;
- employee risk tables;
- department-wise attrition analysis;
- job-role attrition analysis;
- individual employee profile information;
- model-based what-if prediction;
- model performance metrics;
- feature-importance visualization;
- filtered CSV report download; and
- an executive summary section.

The what-if module allows selected values such as OverTime, JobSatisfaction, and WorkLifeBalance to be modified for an employee profile. The system recomputes engineered features, applies the stored preprocessing pipeline, and obtains a new Gradient Boosting probability. This supports scenario exploration without retraining the model.

12. SYSTEM ARCHITECTURE

The system follows a modular architecture:



The trained models and preprocessing object are stored as joblib files. Model performance is stored separately in JSON format. The dashboard loads these saved artifacts and the generated employee risk report rather than retraining the models each time the interface starts.

13. DISCUSSION

The results demonstrate the importance of evaluating several metrics rather than relying on accuracy alone. Gradient Boosting produced the highest test accuracy in this project at 86.05% and a precision of 60.71%, while Logistic Regression produced the highest ROC-AUC at 0.7999 and the highest recall at 61.70%. These differences show that the preferred model can depend on the operational objective.

For an HR risk-screening system, false negatives can be important because a missed high-risk employee may reduce the opportunity for early intervention. At the same time, false positives can create unnecessary attention or intervention. Consequently, the risk score should be treated as a prioritization signal rather than a definitive prediction about an employee's future behavior.

The dashboard therefore emphasizes transparency through metric display, risk categories, employee-level inspection, and what-if analysis. Human review remains necessary before any employment decision is made.

14. LIMITATIONS

The project has several limitations.

1. The dataset is a benchmark-style HR dataset and may not represent the workforce, policies, culture, or labor conditions of a specific organization.
2. The dataset is relatively small for learning complex organizational behavior.
3. The target is imbalanced, and SMOTE can improve training balance without guaranteeing improved real-world generalization.
4. The full employee risk report contains model probabilities generated from the available dataset; these should not be treated as independent prospective predictions.
5. The project uses a single train-test split rather than extensive cross-validation.
6. Risk thresholds of 30% and 60% are project-defined operational thresholds rather than universally validated HR standards.
7. A predicted attrition probability is not a causal explanation of why an employee will leave.
8. Sensitive employment decisions should not be automated solely from model output.

15. FUTURE SCOPE

Future improvements can strengthen both the research and deployment aspects of the system:

- use stratified k-fold cross-validation and repeated evaluation;
- calibrate probability outputs before interpreting them as risk percentages;
- generate out-of-fold or strictly holdout-based risk reports;
- add SHAP-based local and global explanations;
- conduct fairness and subgroup performance analysis;
- compare additional algorithms such as XGBoost, LightGBM, SVM, and calibrated ensembles;
- introduce automated model-drift monitoring;
- connect the dashboard to an HR information system through a secure API;
- add role-based access controls and data privacy safeguards; and
- validate the system on organization-specific longitudinal data before operational deployment.

16. CONCLUSION

This project developed an end-to-end Machine Learning-Based Employee Attrition Prediction and Risk Scoring System. The workflow combines data preprocessing, feature engineering, class-imbalance treatment, supervised learning, model evaluation, probability-based risk scoring, and interactive dashboard visualization.

Three models were evaluated on a held-out test set. The project achieved 86.05% accuracy with Gradient Boosting, while Logistic Regression achieved the highest ROC-AUC among the three at 0.7999 and the highest recall at 61.70%. The final application converts model probabilities into Low, Medium, and High risk categories and presents workforce-level and employee-level analytical views through Streamlit.

The resulting system demonstrates how machine learning can be integrated with HR analytics to create a practical decision-support prototype. Its outputs should be interpreted as analytical signals and used alongside organizational context, human review, and appropriate ethical and privacy safeguards.

17. LEARNING OUTCOMES

Through the project, the following technical and analytical skills were developed:

- Python-based data analysis using Pandas and NumPy.
- Data preprocessing with Scikit-learn.
- Categorical encoding and numerical scaling.
- Feature engineering for HR analytics.
- Handling class imbalance with imbalanced-learn and SMOTE.
- Training and evaluating multiple machine-learning classifiers.
- Saving and loading trained models with Joblib.
- Building interactive dashboards with Streamlit.
- Creating analytical tables and visualizations.
- Generating downloadable CSV reports.
- Structuring a modular project for Git/GitHub.
- Interpreting classification metrics and model limitations.
- Translating machine-learning outputs into an HR analytics interface.

18. REFERENCES

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The literature sources were consulted to position the project methodology within current employee-attrition machine-learning research. The numerical performance values reported in Section 9 are from the present project implementation, not copied from these external studies.

AUTHENTICATION / SIGNATURE

The project report is submitted under the following details:

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